

DATA SCIENCE CAPSTONE PROJECT

Data Science Capstone Project Report

SpaceX Falcon 9 First Stage Landing Prediction

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8 / 21 / 2026

[GitHub Repository](#) ↗

EXECUTIVE SUMMARY

Methods & Key Results at a Glance

METHODS

- › **Data collection** — pulled 90 Falcon 9 launch records from the SpaceX REST API
- › **Web scraping** — parsed 121 historical launches from Wikipedia with BeautifulSoup
- › **Data wrangling** — engineered the binary landing-success label (Class)
- › **EDA** — visual analysis (seaborn/matplotlib) and SQL analysis (SQLite)
- › **Interactive analytics** — a Folium launch-site map and a Plotly Dash dashboard
- › **Machine learning** — tuned 4 classifiers with GridSearchCV to predict landings

90

Falcon 9 launches analyzed

66.7%

historical landing success rate

83.3%

best model test accuracy (SVM)

4

launch sites mapped & compared

BOTTOM LINE

Landing success is predictable from payload, orbit and launch-site data; a tuned SVM classifier reaches 83.3% test accuracy, giving SpaceX's competitors a data-driven way to estimate whether a first stage will be reusable — and therefore how a launch can be priced.

INTRODUCTION

Why Predict a Booster Landing?

BACKGROUND

SpaceX advertises Falcon 9 launches at \$62 million, versus \$165 million+ from other providers. Much of that saving comes from reusing the first-stage booster instead of discarding it in the ocean after every flight.

If we can predict whether a first stage will land successfully, we can estimate the true cost of a launch — information a competitor could use when bidding against SpaceX.

PROBLEM STATEMENT

Predict whether the Falcon 9 first stage will land successfully, using historical launch data.

PROJECT PIPELINE

- 1 Collect** — SpaceX API + Wikipedia scraping
- 2 Wrangle** — build the landing-success label
- 3 Explore** — EDA with charts, SQL, maps & dashboard
- 4 Predict** — tune & compare 4 ML classifiers

Pulling Launch Records from the REST API

PIPELINE

- 1** **GET /launches/past** — fetch a consistent static snapshot of every past launch as JSON
- 2** **json_normalize()** — flatten the response into a Pandas DataFrame
- 3** **Filter single-core/payload** — keep flights with exactly one core and one payload
- 4** **Enrich by ID** — call /rockets, /launchpads, /payloads, /cores for each launch
- 5** **Keep Falcon 9 only** — drop Falcon 1 flights, renumber FlightNumber
- 6** **Impute & export** — fill missing PayloadMass with the mean → dataset_part_1.csv

KEY PHRASES

REST API

JSON normalize

ID enrichment

single-core filter

mean imputation

CSV export

NETWORK NOTE

The notebook demonstrates the live per-ID enrichment calls, then transparently falls back to SpaceX's official pre-collected snapshot of this exact dataset if `api.spacexdata.com` is unreachable — exactly as the assignment's own instructions recommend for downstream consistency.

[View notebook on GitHub ↗](#)

Parsing Launch History from Wikipedia

PIPELINE

- 1 Request fixed revision** — GET a pinned Wikipedia snapshot (9 Jun 2021) for reproducibility
- 2 Parse with BeautifulSoup** — load the HTML response into a parse tree
- 3 Extract table headers** — clean each `<th>` into a column name
- 4 Walk every launch table** — iterate `<tr>/<td>` rows across all tables on the page
- 5 Clean noisy cells** — strip refs/footnotes from date, mass, orbit & outcome fields
- 6 Export** — build a DataFrame of 121 records → `spacex_web_scraped.csv`

KEY PHRASES

BeautifulSoup

HTML table parsing

regex cleanup

unicode normalize

121 records

RESULT

Live-scraped directly against Wikipedia in this run: 121 launch records with flight number, date/time, booster version, launch site, payload, orbit, customer, and outcome — a second, independent source that cross-checks the API-collected dataset.

[View notebook on GitHub ↗](#)

From Raw Outcomes to a Training Label

Every launch record includes an Outcome string describing how (or whether) the booster landed — e.g. True ASDS, False Ocean, None None. Wrangling turns that free-text outcome into a clean binary label the ML models can learn from.

1

Missing values

Checked % missing per column;
LandingPad keeps None (= no pad used)
as a real category, not an error.

2

Class label

True Ocean / RTLS / ASDS → Class = 1
(success). False..., None ASDS, None
None → Class = 0 (failure).

3

Imputation

Missing PayloadMass values replaced
with the column mean so no launch is
dropped.

4

Export

Result saved as dataset_part_2.csv — the
base table for every EDA and ML step that
follows.

RESULT

90 Falcon 9 launches · 66.7% historical landing success rate

[View notebook on GitHub ↗](#)

Charts Used & What They Reveal

Chart	Variables	Purpose
Categorical scatter	Flight Number × Payload Mass (hue = outcome)	does experience/payload affect landing?
Categorical scatter	Flight Number × Launch Site	how site usage evolved over time
Categorical scatter	Payload Mass × Launch Site	per-site payload capacity limits
Bar chart	Success rate by Orbit	which orbits are hardest to land from
Categorical scatter	Flight Number × Orbit	orbit adoption over the program's history
Categorical scatter	Payload Mass × Orbit	payload feasibility per orbit type
Line chart	Year × average success rate	the year-over-year improvement trend

FEATURE ENGINEERING

Categorical columns (Orbit, LaunchSite, LandingPad, Serial) are one-hot encoded with `pd.get_dummies()` and cast to `float64`, producing the ML-ready feature table `dataset_part_3.csv`.

[View notebook on GitHub ↗](#)

Ten Questions Answered with SQLite

#	Question
1	Unique launch sites used in the mission history
2	First 5 records where the launch site begins with 'CCA'
3	Total payload mass carried for NASA (CRS) missions
4	Average payload mass carried by booster version F9 v1.1
5	Date of the first successful ground-pad landing
6	Boosters that succeeded on a drone ship with 4,000–6,000 kg payload
7	Total successful vs. failed mission outcomes
8	Booster versions that carried the maximum payload mass (subquery)
9	2015 drone-ship landing failures by month, booster & site
10	Landing outcomes ranked by count, 2010-06-04 to 2017-03-20

Folium Map & Plotly Dash Dashboard

FOLIUM MAP

- › All 4 launch sites plotted with circle markers and labels
- › Every historical launch clustered on the map — green marker for a success, red for a failure
- › MousePosition tool + haversine distance formula measure proximity to coastline, access road & nearest city

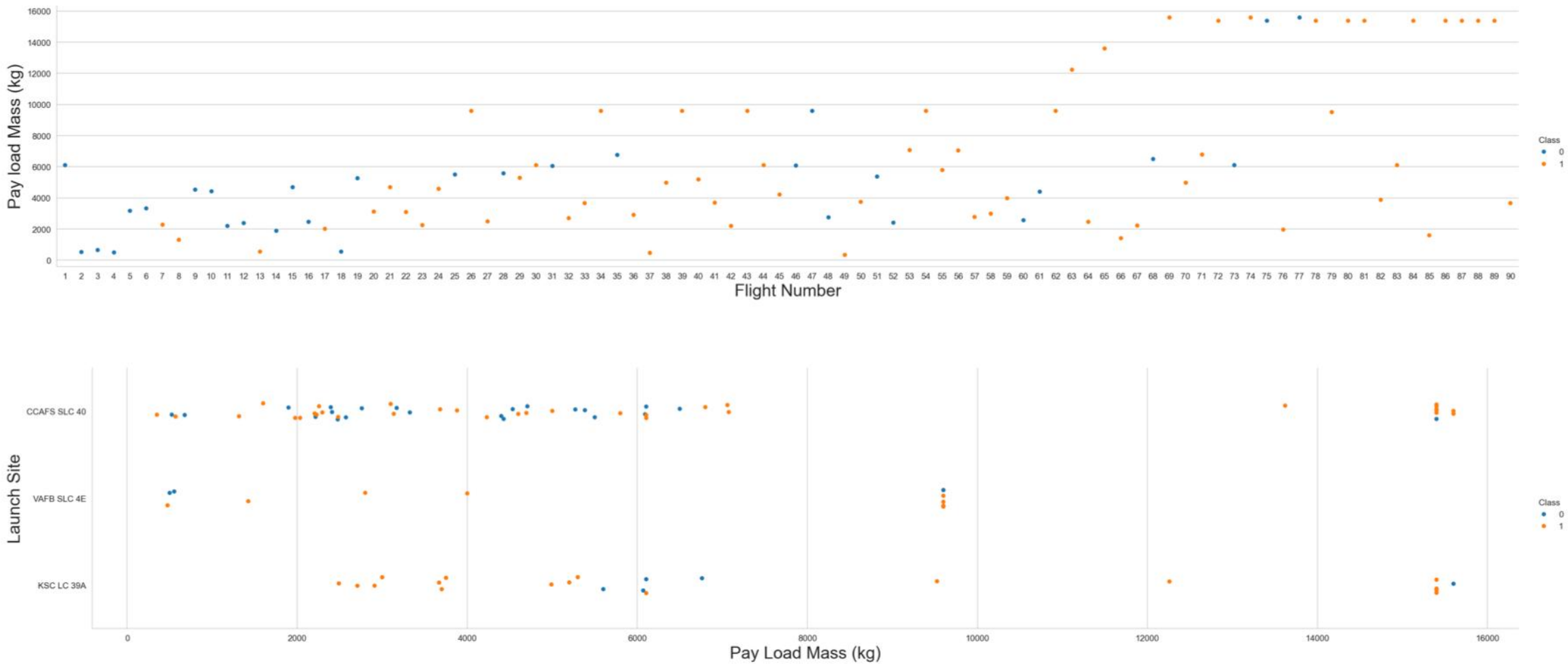
[View notebook on GitHub ↗](#)

PLOTLY DASH DASHBOARD

- › Dropdown to filter by launch site (or view all four together)
- › Pie chart of success vs. failure counts, updating live with the dropdown
- › Range slider (0–10,000 kg) driving a payload-vs-outcome scatter plot, colored by booster version

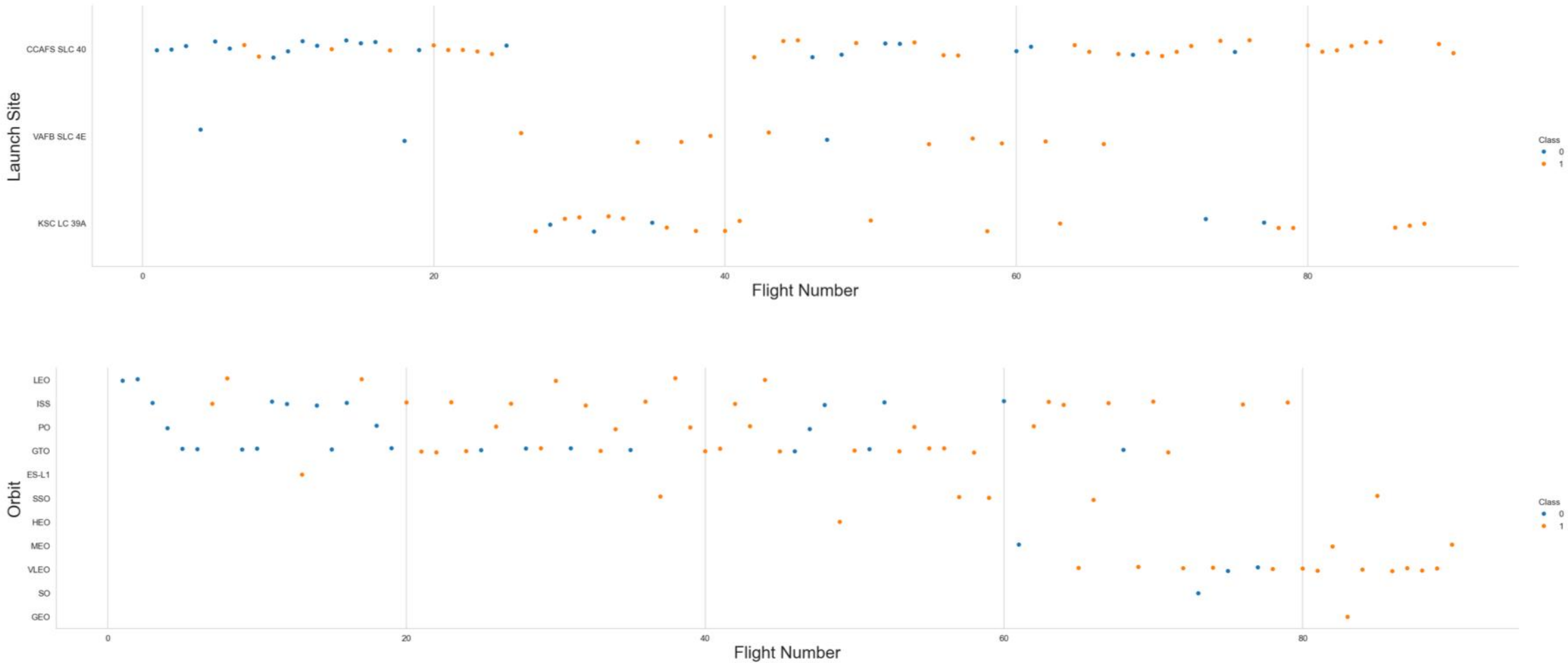
[View app on GitHub ↗](#)

Payload Mass vs. Flight Number & Launch Site



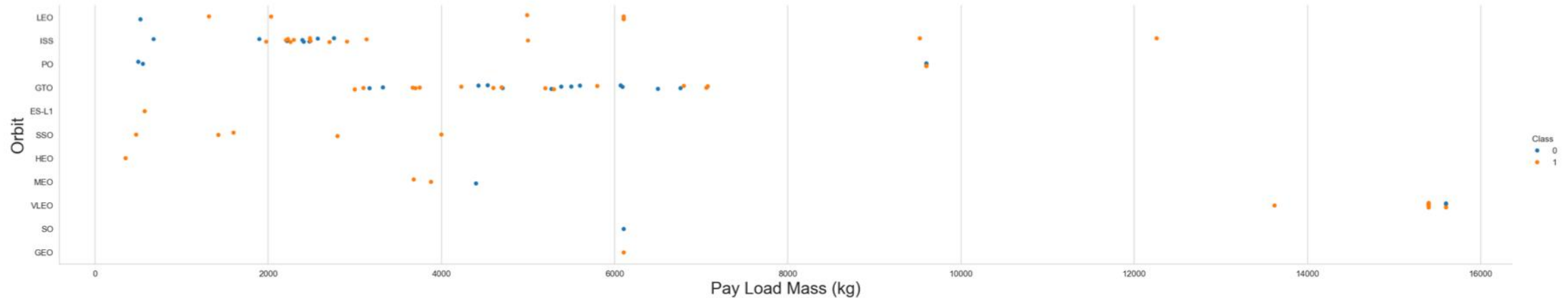
As flight number grows the first stage lands more reliably; VAFB SLC-4E never launches payloads over ~10,000 kg.

Launch Site & Orbit vs. Flight Number



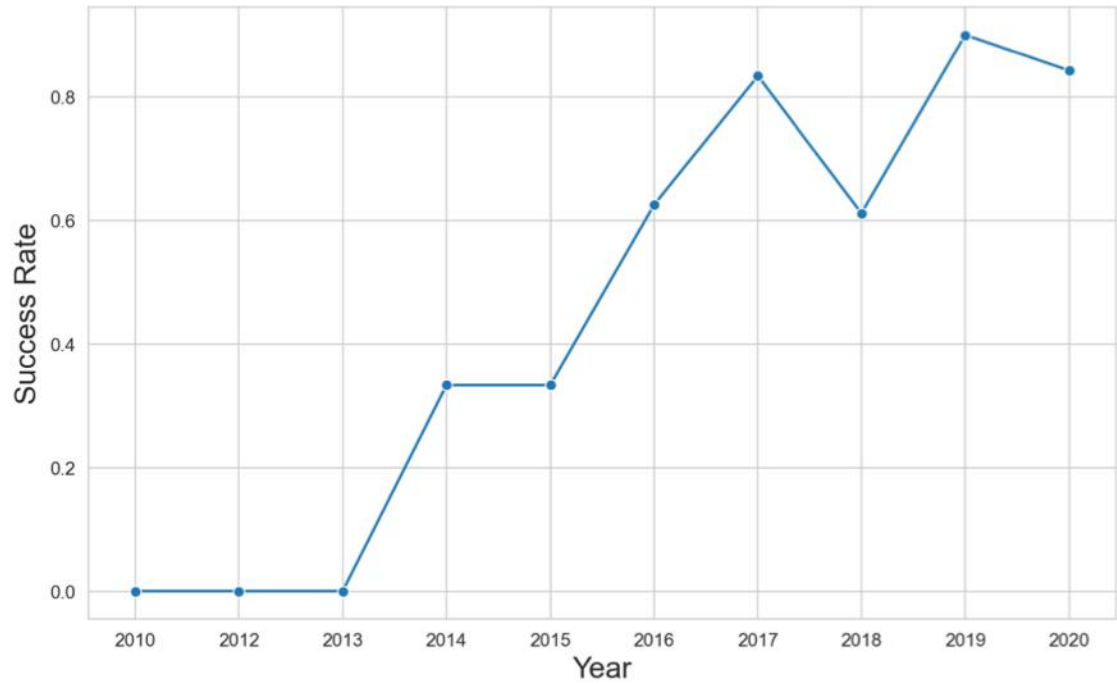
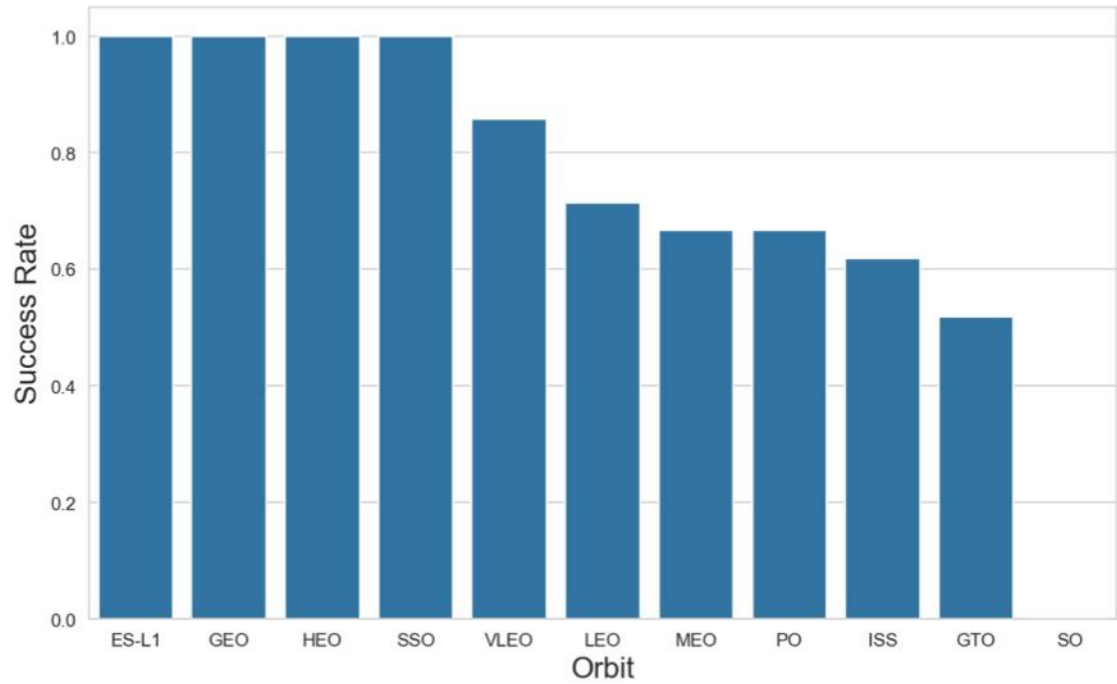
KSC LC-39A and CCAFS SLC-40 host the more recent, higher-success flights; in LEO, success tracks flight number — in GTO it doesn't.

Which Orbits Tolerate Heavy Payloads?



With heavy payloads, Polar, LEO and ISS orbits show mostly successful landings. For GTO, both outcomes appear across the full payload range — orbit alone doesn't determine success there.

Bar Chart and Yearly Trend



Launch Sites & Payload Mass

Q1 · UNIQUE LAUNCH SITES

CCAFS LC-40

CCAFS SLC-40

KSC LC-39A

VAFB SLC-4E

Q2 · FIRST 5 RECORDS AT 'CCA%' SITES

Date	Booster	Payload (kg)
2010-06-04	B0003	0
2010-12-08	B0004	0
2012-05-22	B0005	525
2012-10-08	B0006	500
2013-03-01	B0007	677

Q3 · TOTAL NASA (CRS) PAYLOAD

45,596 kg

Q4 · AVERAGE PAYLOAD, BOOSTER F9 v1.1

2,534.67 kg

Q8 · MAXIMUM PAYLOAD EVER CARRIED

15,600 kg

— tied among 12 different F9 B5 boosters

[Run these queries on GitHub ↗](#)

Landing Milestones & Outcome Rankings

Q5 · FIRST SUCCESSFUL GROUND-PAD LANDING

2015-12-22

Q7 · MISSION OUTCOMES

Mission Outcome	Count
Success	99
Failure (in flight)	1
Success (payload status unclear)	1

Q6 · DRONE-SHIP SUCCESSES, 4,000–6,000 KG PAYLOAD

F9 FT B1021.2 · B1022 · B1026 · B1031.2

Q10 · LANDING OUTCOMES RANKED, 2010-06-04 → 2017-03-20

Landing Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

2015 Drone-Ship Failures — Time Analysis

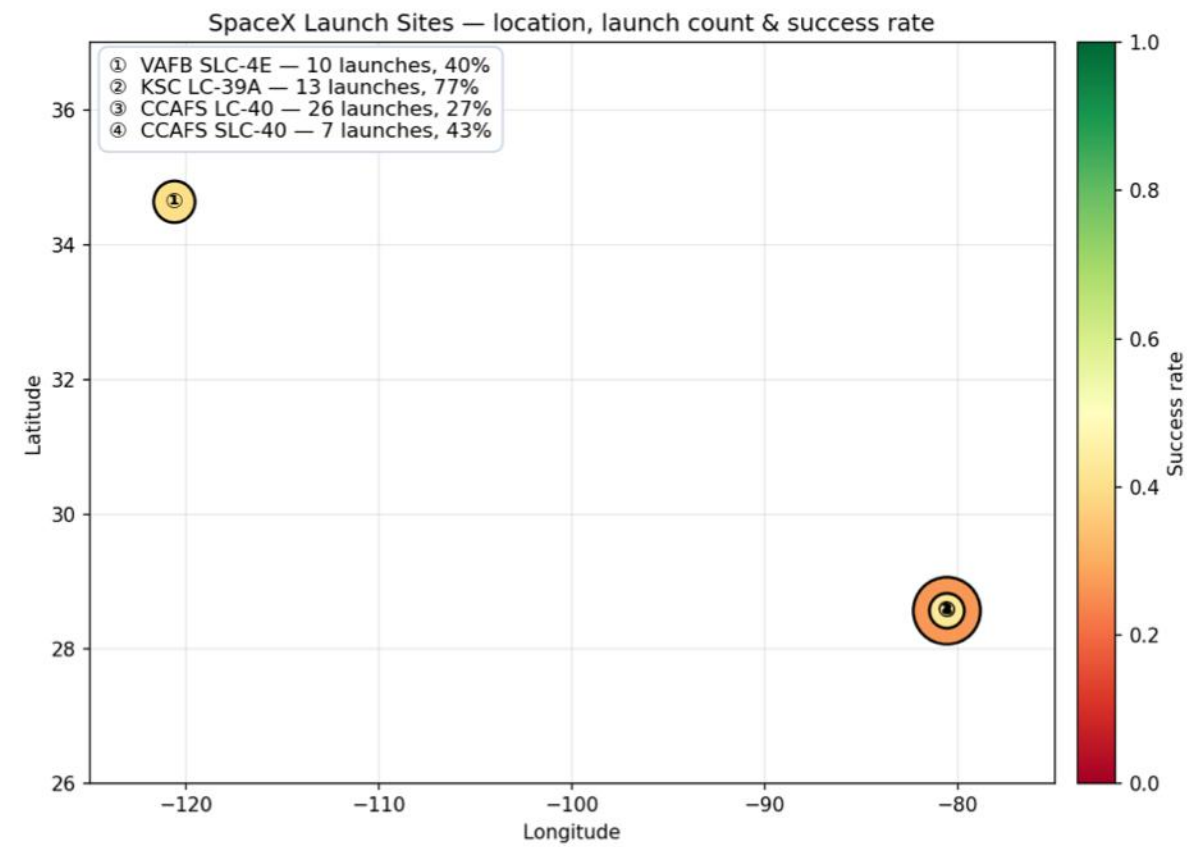
Q9 · MONTH, BOOSTER & SITE FOR EVERY 2015 DRONE-SHIP FAILURE

Month	Landing Outcome	Booster Version	Launch Site
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

READING THE RESULT

Both 2015 drone-ship attempts failed, and both flew the earlier F9 v1.1 booster from the same CCAFS LC-40 pad — well before SpaceX's first successful ground-pad landing that December. This SQL query pinpoints exactly when and on what hardware the drone-ship landing technique was still unproven, which lines up with the rising success-rate trend seen in the EDA charts from 2013 onward.

Launch Sites & Launch Records



Launch Site	Launches	Success rate
KSC LC-39A	13	76.9%
CCAFS SLC-40	7	42.9%
VAFB SLC-4E	10	40.0%
CCAFS LC-40	26	26.9%

MARKER CLUSTERS

Every one of the 90 API-collected launches is plotted at its exact site coordinates, colored green for a successful landing and red for a failure, and grouped into clusters that expand as you zoom in — so the full mission history is visible at a glance.

Marker size = number of launches · color = success rate (red → green)

Proximity Analysis — CCAFS SLC-40

Using the map's MousePosition readout and the haversine formula, we measured the straight-line distance from the launch pad to its nearest coastline, access road, and town:

0.51 km

to the nearest coastline

0.58 km

to Samuel C. Phillips Pkwy (highway)

17.72 km

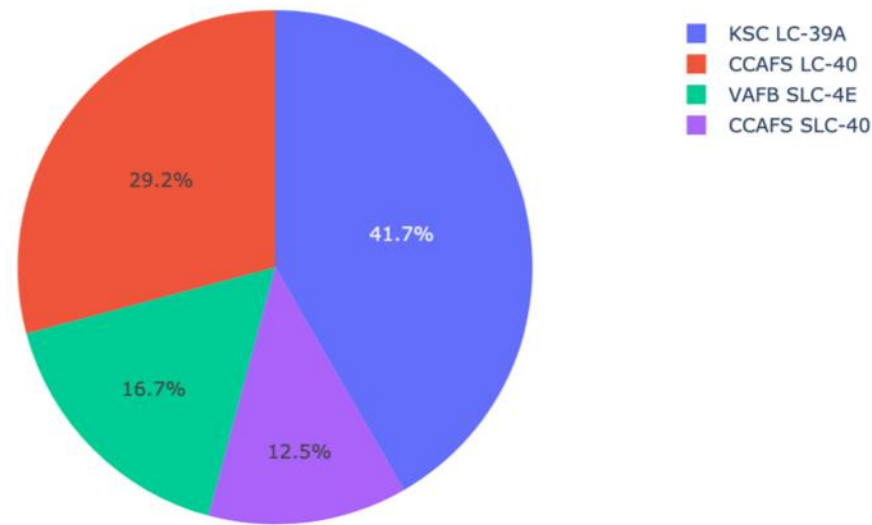
to Cape Canaveral, FL (nearest city)

FINDINGS

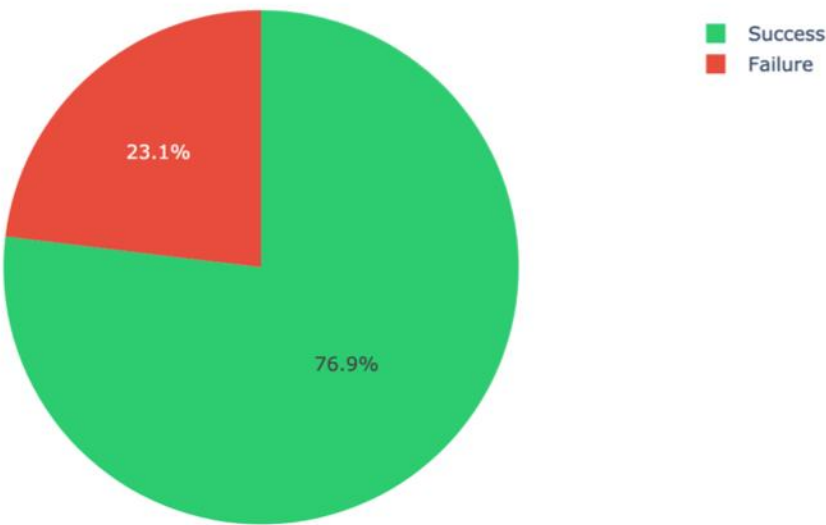
Launch pads sit under 1 km from both the ocean and their access road — close enough for efficient logistics — while staying over 17 km from the nearest town, keeping the public well clear of the blast radius. Every SpaceX site follows this same coastal, low-latitude pattern for the same reasons: safety for failed launches and a rotational-speed boost from being near the equator.

Success Pie Charts

Total Successful Launches by Site

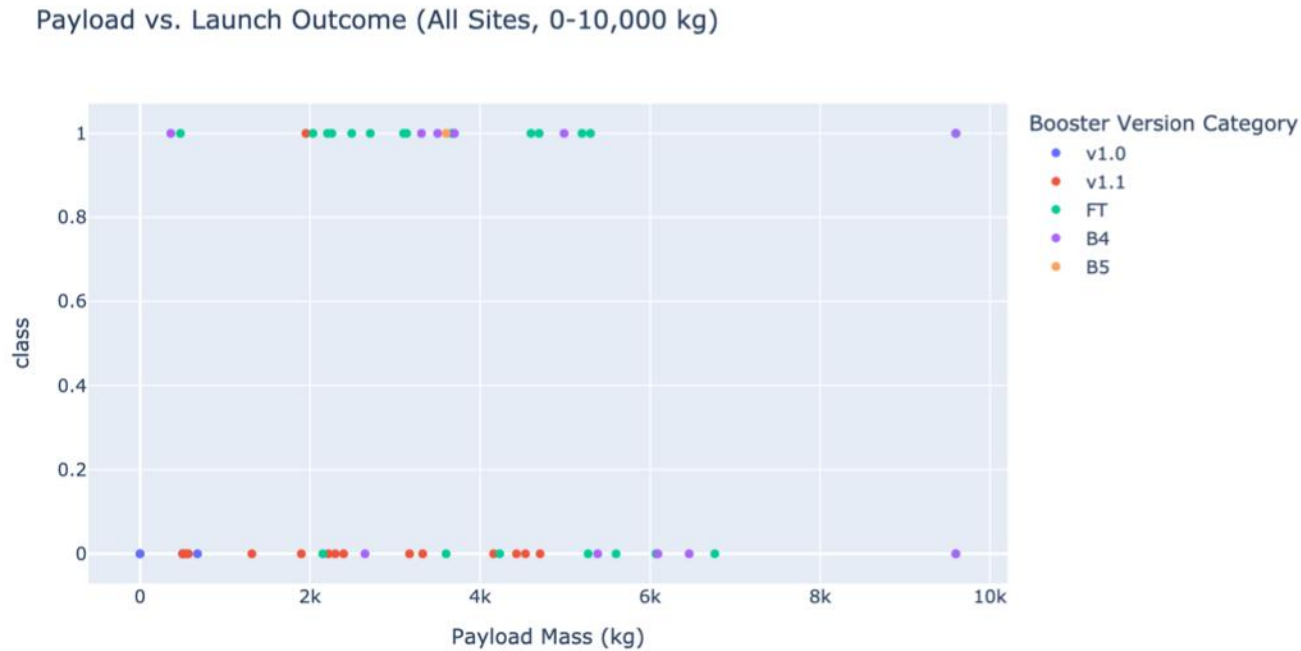


Success vs. Failure for site KSC LC-39A



KSC LC-39A leads on both counts: it has the most successful launches (10) and the highest success rate (76.9%) of any site.

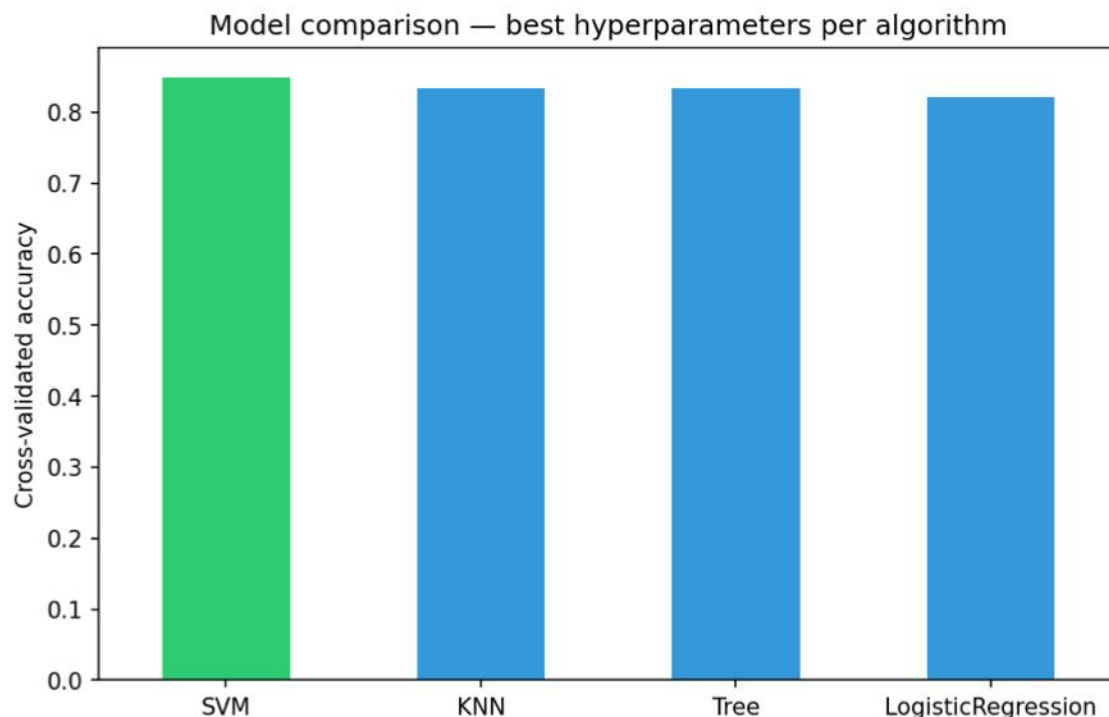
Payload vs. Launch Outcome



DASHBOARD INSIGHTS

- › 2,000–4,000 kg payloads land most reliably (61.9% success)
- › 6,000–8,000 kg payloads never landed successfully in this dataset (0%)
- › Booster version B5 has a perfect 100% landing success rate

Four Classifiers, Tuned with GridSearchCV



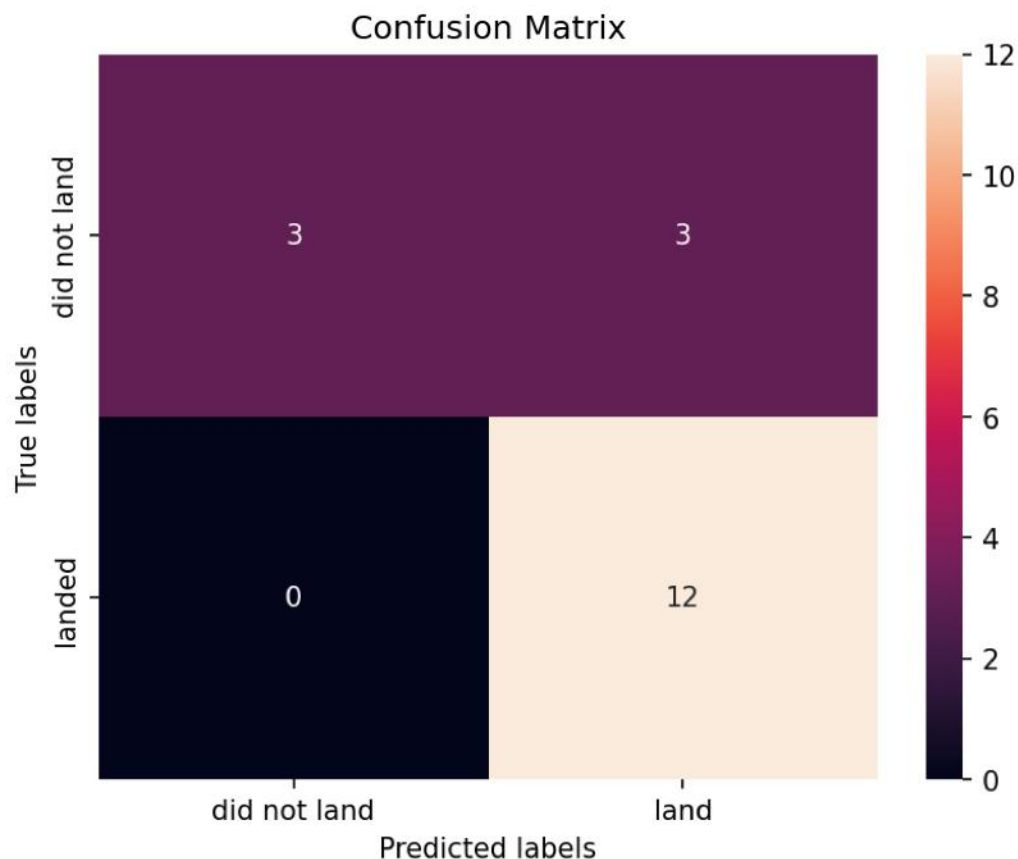
Model	CV Acc.	Test Acc.	Best hyperparameters
SVM	84.8%	83.3%	kernel=sigmoid, C=1, $\gamma=0.032$
KNN	83.4%	83.3%	k=6, p=1, algorithm=auto
Decision Tree	83.4%	72.2%	gini, max_depth=10
Logistic Reg.	82.1%	83.3%	C=1, penalty=l2, lbfgs

BEST MODEL

Support Vector Machine

Highest cross-validated accuracy (84.8%) among all four tuned models, with a sigmoid kernel found via grid search over kernel, C and gamma.

SVM Confusion Matrix & Evaluation



READING THE CONFUSION MATRIX

- › **Recall on landings: 100%** — the model correctly caught all 12 successful landings in the 18-launch test set.
- › **False positives: 3** — its only errors were predicting a landing on 3 flights that actually failed.
- › **False negatives: 0** — it never missed a real successful landing.
- › **Test accuracy: 83.3%** — 15 of 18 held-out launches correctly classified.

For a competitor estimating launch cost, this bias toward optimism (never missing a real landing, occasionally over-predicting one) is the safer failure mode — it avoids underestimating SpaceX's reusability advantage.

[View notebook on GitHub ↗](#)

CONCLUSION & INSIGHTS

What We Learned — and What's Next

- › **Landing success is learnable.** Payload mass, orbit, launch site and flight number carry enough signal for an 83.3% accurate predictor.
- › **Experience compounds.** Success rate climbed from near-zero in 2013 to over 80% by 2020 as SpaceX iterated on the landing process.
- › **Site and orbit matter as much as hardware.** KSC LC-39A (76.9%) and 2,000–4,000 kg / LEO-ISS payloads consistently out-land the rest.



A CREATIVE NEXT STEP

Build a "reusability index" — combining Block version, ReusedCount and launch-site success history into a single feature — and feed it back into the SVM. Paired with the Dash dashboard, a competitor could quote a real-time, per-mission cost estimate the moment a customer specifies payload mass and target orbit, instead of relying on SpaceX's published price alone.

Bottom line: a data-driven pipeline — from a public API and Wikipedia to a tuned SVM — turns SpaceX's reusability advantage into a number any competitor can act on.



Thank you

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github.com/MissBiancaCh/Data-Science-Capstone-Project-Report